PCBC H Name _____ _____ /32

Limits Quiz 1.1–1.2

Non-Calculator

You have 30 minutes.

PRACTICE A · built to match the real Quiz 1.1–1.2 (same layout, same question types)

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

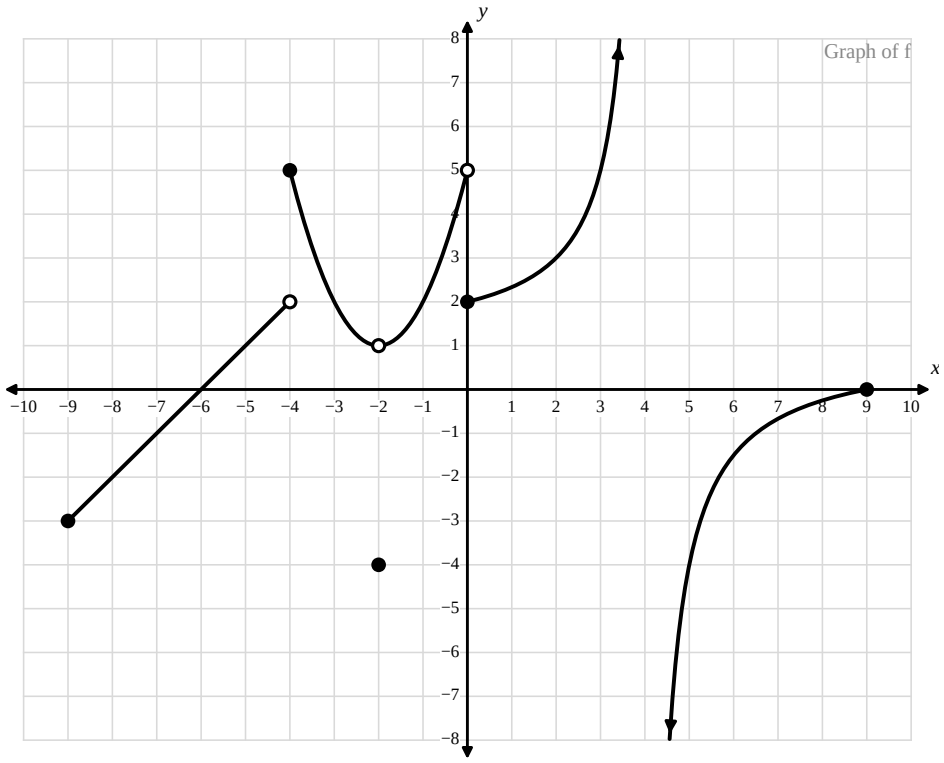
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Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (1 point each)

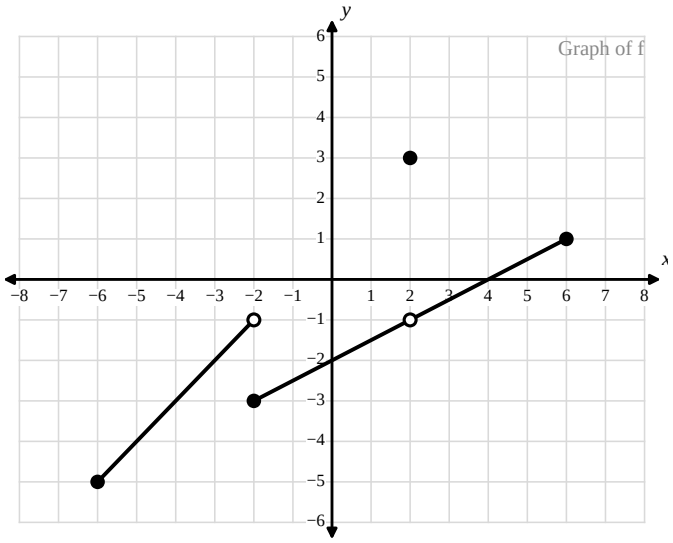


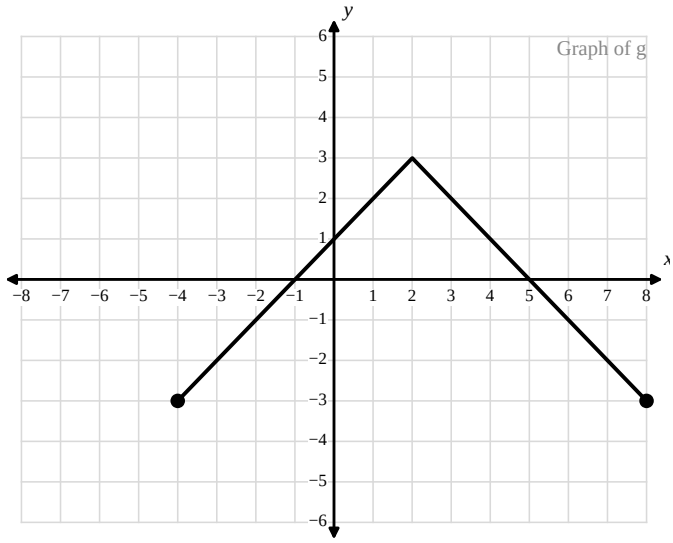
1. $\lim_{x\to -2} f(x) =$	**2.** $f(-2) =$
3. $\lim_{x\to -4} f(x) =$	**4.** $\lim_{x\to 4^+} f(x) =$

(2 points)

5. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = 2$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

Use the graphs of f and g below to find each limit. Show the limit properties you use. (2 points each)





6. $\lim_{x \to 2} [f(x)\, g(x)]$

7. $\lim_{x \to -2^+} [3f(x) - g(x)]$

(2 points)

8. Use the table to estimate $\lim_{x \to 3} g(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	2.9	2.99	2.999	$\to 3 \leftarrow$	3.001	3.01	3.1
$g(x)$	0.2515823	0.2501564	0.2500156		0.2499844	0.2498439	0.2484567

$\lim_{x \to 3^-} g(x) \approx$ _____ $\lim_{x \to 3^+} g(x) \approx$ _____ $\lim_{x \to 3} g(x) \approx$ _____

Evaluate each limit. (2 points each)

9. $\lim_{x \to -2} \left(3x^3 - x^2 + 4\right)$	**10.** $\lim_{x \to \frac{\pi}{3}} \left(2\sec x - \tan x\right)$
11. $\lim_{x \to 2} f(x), \quad f(x) = \begin{cases} x^2 - 3, & x < 2 \\ 3x - 5, & x \geq 2 \end{cases}$	**12.** $\lim_{x \to 3^-} \dfrac{x - 3}{\lvert x - 3 \rvert}$

Evaluate each limit. (3 points each)

13. $\displaystyle\lim_{x\to -3}\frac{x^2+x-6}{x^2-9}$

14. $\displaystyle\lim_{x\to 2}\frac{x-2}{\sqrt{2x+5}-3}$

15. $\displaystyle\lim_{x\to 0}\frac{\frac{1}{x+3}-\frac{1}{3}}{x}$

16.

For $f(x)=x^2-5x+2$, find $\displaystyle\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$

Each step: the work on the left, the reason on the right. "lim" stays on every line until the line where you substitute.

1. $\boxed{1}$

Both sides of the curve head to the open circle at $(-2, 1)$.	The filled dot at $(-2, -4)$ is $f(-2)$; a limit ignores the point at $x = -2$.

2. $\boxed{-4}$

The filled dot at $x = -2$ is at height -4.	Function value: use the filled dot, not the hole.

3. $\boxed{\text{DNE}}$

$\lim_{x \to -4^-} f(x) = 2$, $\lim_{x \to -4^+} f(x) = 5$	From the left the line ends at the open circle $(-4, 2)$; from the right the curve starts at $(-4, 5)$.
$2 \neq 5$, so $\lim_{x \to -4} f(x)$ does not exist.	Two-sided needs both sides to match.

4. $\boxed{-\infty}$

Just right of $x = 4$ the curve falls without bound.	One-sided limit from the right only. (From the left it rises to $+\infty$.)

5. $\boxed{a = -3,\ a = -1}$

The curve reaches height 2 from both sides at $x = -3$ and $x = -1$.	Two-sided: approach $y = 2$ from the left and the right.
Not $x = -4$: only the left side heads to 2 (the right side starts at 5).	A one-sided approach does not count.
Not $x = 0$: only the right side starts at 2 (the left side heads to 5).	Same reason.

6. $\boxed{-3}$

$\lim_{x \to 2} [f(x)g(x)] = \left[\lim_{x \to 2} f(x)\right]\left[\lim_{x \to 2} g(x)\right]$	Limit of a product. Write the property first.
$= (-1)(3)$	f has a hole at $(2, -1)$ (ignore the dot at $(2, 3)$); g peaks at $(2, 3)$.
$= -3$	Multiply.

7. $\boxed{-8}$

$\lim_{x \to -2^+} [3f(x) - g(x)] = 3 \lim_{x \to -2^+} f(x) - \lim_{x \to -2^+} g(x)$	Constant multiple and difference properties.
$= 3(-3) - (-1)$	From the right, f starts at the filled dot $(-2, -3)$; $g(-2) = -1$.
$= -8$	Trap: the left side of f heads to -1, which would give -2.

8. $\boxed{\approx 0.25}$

Left: 0.2515823, 0.2501564, 0.2500156	Closest value rounds to 0.250.
Right: 0.2484567, 0.2498439, 0.2499844	Closest value rounds to 0.250.
$\lim_{x \to 3} g(x) \approx 0.25$	Both sides agree. A table estimates; it does not prove.

9. $\boxed{-24}$

$\lim_{x\to -2}\left(3x^3-x^2+4\right)=3(-2)^3-(-2)^2+4$	Polynomial: substitute.
$=-24-4+4=-24$	Simplify.

10. $\boxed{4-\sqrt{3}}$

$\lim_{x\to \frac{\pi}{3}}\left(2\sec x-\tan x\right)=2\sec\dfrac{\pi}{3}-\tan\dfrac{\pi}{3}$	$\pi/3$ is in the domain: substitute.
$=2(2)-\sqrt{3}=4-\sqrt{3}$	$\sec\frac{\pi}{3}=\frac{1}{\cos(\pi/3)}=2$, $\tan\frac{\pi}{3}=\sqrt{3}$.

11. $\boxed{1}$

$\lim_{x\to 2^-} f(x)=\lim_{x\to 2^-}\left(x^2-3\right)=4-3=1$	Left of 2 uses x^2-3.
$\lim_{x\to 2^+} f(x)=\lim_{x\to 2^+}(3x-5)=6-5=1$	Right of 2 uses $3x-5$.
$\lim_{x\to 2} f(x)=1$	$1=1$: the sides match.

12. $\boxed{-1}$

$\lim_{x\to 3^-}\dfrac{x-3}{\lvert x-3\rvert}=\lim_{x\to 3^-}\dfrac{x-3}{-(x-3)}$	For $x<3$, $x-3$ is negative, so $\lvert x-3\rvert=-(x-3)$.
$=\lim_{x\to 3^-}(-1)=-1$	Cancel; limit of a constant.

13. $\boxed{\frac{5}{6}}$

Substitution produces 0/0: indeterminate.	Factor.
$\lim_{x\to -3}\dfrac{x^2+x-6}{x^2-9}=\lim_{x\to -3}\dfrac{(x+3)(x-2)}{(x+3)(x-3)}$	Trinomial and difference of squares.
$=\lim_{x\to -3}\dfrac{x-2}{x-3}$	Cancel $x+3$; allowed because $x\neq -3$.
$=\dfrac{-3-2}{-3-3}=\dfrac{-5}{-6}=\dfrac{5}{6}$	Substitute.

14. $\boxed{3}$

Substitution produces 0/0: indeterminate.	Radical: multiply by the conjugate.
$\lim_{x\to 2}\dfrac{x-2}{\sqrt{2x+5}-3}=\lim_{x\to 2}\dfrac{(x-2)\left(\sqrt{2x+5}+3\right)}{(2x+5)-9}$	Bottom is a^2-b^2; leave the top factored.
$=\lim_{x\to 2}\dfrac{(x-2)\left(\sqrt{2x+5}+3\right)}{2(x-2)}$	$2x-4=2(x-2)$.
$=\lim_{x\to 2}\dfrac{\sqrt{2x+5}+3}{2}$	Cancel $x-2$.
$=\dfrac{\sqrt{9}+3}{2}=\dfrac{6}{2}=3$	Substitute.

15. $\boxed{-\frac{1}{9}}$

Substitution produces 0/0: indeterminate.	Complex fraction: multiply top and bottom by the LCD $3(x+3)$.
$\lim\limits_{x\to 0}\dfrac{\frac{1}{x+3}-\frac{1}{3}}{x}=\lim\limits_{x\to 0}\dfrac{3-(x+3)}{3x(x+3)}$	$3(x+3)\cdot\frac{1}{x+3}=3$ and $3(x+3)\cdot\frac{1}{3}=x+3$.
$=\lim\limits_{x\to 0}\dfrac{-x}{3x(x+3)}$	Distribute the negative: $3-x-3=-x$.
$=\lim\limits_{x\to 0}\dfrac{-1}{3(x+3)}$	Cancel x.
$=\dfrac{-1}{3(0+3)}=-\dfrac{1}{9}$	Substitute.

16. $\boxed{2x-5}$

$\lim\limits_{h\to 0}\dfrac{\left[(x+h)^2-5(x+h)+2\right]-\left(x^2-5x+2\right)}{h}$	Build $f(x+h)$ with brackets.
$=\lim\limits_{h\to 0}\dfrac{x^2+2xh+h^2-5x-5h+2-x^2+5x-2}{h}$	Expand; distribute the negative.
$=\lim\limits_{h\to 0}\dfrac{h(2x+h-5)}{h}$	Combine; factor out h.
$=\lim\limits_{h\to 0}(2x+h-5)=2x+0-5=2x-5$	Cancel h; substitute 0 for h, not x.